

Institute of Actuaries of India

Subject CM2A – Financial Engineering and Loss Reserving (Paper A)

March 2021 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:

- i) B
Reason: I and iii are correct & II is incorrect since higher Initial wealth would result in lower Probability of ruin [2]
- ii) A
I and II are attributes of overconfidence [2]
- iii) C
Except C, all the other 3 are components of the prospect theory [2]
- iv) B
Main cause for the equity puzzle is “Myopic loss aversion” [2]
- v) C
The fact that we can predict the future based on statistical analysis contradicts Weak Form of EMT. Insiders can profit from insider trading does not tell anything about contradiction of Weak Form of EMT. [2]

[10 Marks]**Solution 2:**

- I. Adverse Selection
II. Adverse Selection
III. Moral Hazard
IV. Moral Hazard

[2 Marks]**Solution 3:**

Event Category	Risk incident	Action
Low-frequency Low-severity	An employee getting locked in the office restroom for a few minutes	Ignore
High-frequency Low-severity	Undetected shoplifting at retail store e.g. a biscuit packet opened and eaten in the store	Control
Low-frequency High-severity	Fire at the head office of a life insurance company	Insure
High-frequency High-severity	An earthquake striking a nuclear reactor situated in an earthquake-prone area	Avoid

[0.5 mark for each correct match and 0.5 mark for each correct action]

[4 Marks]**Solution 4:**

- i)
- Value at risk (VaR) represents the maximum potential loss on a portfolio over a given future time period with a given degree of confidence (1-p).
 - The expected shortfall is $E[\max(L-X, 0)]$
[0.5 each – either formula or explanation in words is acceptable]
 - VaR is not sub-additive whereas expected shortfall is sub-additive i.e. on adding 2 risk portfolios, their combined VaR may increase or decrease compared to the sum of individual VaRs, but expected shortfall can never increase.

[1]

The other similar measures are:

- Expected tail loss

- Tail conditional expectation
- Conditional VaR
- tail conditional VaR
- worst conditional expectation

[0.5 each, max 2]

[4]

ii)

Shortfall probability = probability that the investment will be worth less than 25 lakh after 5 years

$$= \Pr(1950000S_5 < 2500000)$$

[1]

So the required probability is $\Pr(\ln(S_5) < \ln(2500000/1950000)) = \Pr(\ln(S_0) + (\mu - \sigma^2/2)t + \sigma Z_t < \ln(1.282051)) = \Pr(0.3360 + 0.16Z_5 < 0.248461)$ (... substituting $t = 5$, and the values of S_0 , μ and σ)

[1]

We know that Z_5 is normally distributed with mean 0 and variance = 5. So Z_5 can be written as $\sqrt{5}z$ where z is a standard normal random variable.

[1]

The required probability = $\Pr(0.3578z < -0.08754) = \Pr(z < -0.24468) = 0.403353$

Therefore, there is a 40.3% probability that Mr. Tourist's investment will fall short of the target.

[1]

We know S_t follows a lognormal distribution with parameters $(\mu - \sigma^2/2)t$ and $\sigma\sqrt{t}$.

$$\text{So, } E[S_t] = \exp((\mu - \sigma^2/2)t + (\sigma^2 t/2)) = \exp(\mu t).$$

[1]

The expected value of investment of 19.5 lakh after 5 years is $19,50,000 \cdot \exp(0.08 \cdot 5) = \text{Rs. } 29,09,058$.

[1]

[6]**[10 Marks]****Solution 5:**i) $(1+i) \sim \log N(\mu, \sigma^2)$

$$E(i) = .1$$

$$V(i) = .04^2$$

$$E(1+i) = 1 + E(i) = 1.1$$

$$e^{(u+1/2\sigma^2)} = 1.1$$

[1]

$$V(1+i) = V(i) = .04^2$$

$$e^{2u+\sigma^2} (e^{\sigma^2} - 1) = .04^2$$

[1]

Squaring equation 1:

$$e^{2u+\sigma^2} = 1.1^2$$

Substituting in equation 2

$$1.1^2(e^{\sigma^2} - 1) = .04^2$$

[1]

$$\sigma^2 = \ln[(.04^2 / 1.1^2) + 1] = 0.001321$$

$$\mu = \ln(1.1) - \sigma^2/2 = 0.094649$$

[1]

[4]

- ii) Let S_{10} denote the accumulated value of 1 at time 10.

Let X be the amount invested at time 0. Accumulated value of fund after 10 years is XS_{10}

Since yields are independent,

$$S_{10} \sim \log N(10\mu, 10\sigma^2) = \log N(0.94649, 0.01321) \quad [1]$$

$$P(XS_{10} \geq 200000) = P(S_{10} \geq 200000/X)$$

$$= P(\ln S_{10} \geq \ln(200000/X))$$

$$= P(Z \geq \{\ln(200000/X) - 0.94649\} / [0.01321^{(0.5)}]) \quad [1]$$

Since this probability is 0.99, $\{\ln(200000/X) - 0.94649\} / [0.01321^{(0.5)}]$ must be lower 1% of the standard normal distribution, i.e:

$$\{\ln(200000) - \ln(X) - 0.94649\} / [0.01321^{(0.5)}] = -2.3263 \quad [1]$$

Solving for X

$$\{\ln(200000) - \ln(X) - 0.94649\} = -2.3263 * 0.01321^{(0.5)}$$

$$\ln(X) = \ln(200000) + 0.267373 - 0.94649 = 11.52696 \quad [1]$$

$$X = e^{11.52696} = 101413.3 \quad [1]$$

[5]

[9 Marks]

Solution 6:

- i) Answer is b. [2]

The probability that the bond for company A does not default between times 5 and 10 is:

$$\text{Exp} \left\{ -\int_5^{10} \lambda(s) ds \right\} = \text{Exp} \left\{ -\int_5^{10} .05s ds \right\} = 0.1534$$

- ii)

The probability that the company will default by time 10 is:

$$\begin{aligned} P(10) &= 1 - \text{Exp} \left\{ -\int_0^{10} \lambda(s) ds \right\} = 1 - \text{Exp} \left\{ -\int_0^{10} 0.0015s^2 + 0.006s ds \right\} \\ &= 1 - \text{Exp} \left\{ -(0.0005s^3 + .003s^2)_0^{10} \right\} \\ &= 1 - \text{Exp} \{ - (0.8) \} = 0.5507 \end{aligned} \quad [2]$$

- iii) The general formula for the price of a zero-coupon bond under the two state model for credit risk using a risk-neutral probability measure is:

$$v(t, T) = e^{-r(T-t)} [1 - (1-\delta) \{1 - \exp(-\int_t^T \lambda(s) ds)\}] \quad [1]$$

For Bond issued by company A

$$\begin{aligned} B_A &= e^{-0.05 * 10} [1 - (1-\delta) \{1 - \exp(-\int_0^{10} .05s ds)\}] \\ B_A &= e^{-0.5} [1 - (1-\delta) \{1 - \exp(-2.5)\}] \end{aligned}$$

[0.5]

For Bond issued by company B

$$B_B = e^{-0.05 * 10} [1 - (1-\delta) \{1 - \exp(-\int_0^{10} .0015s^2 + 0.006s ds)\}]$$

$$B_B = e^{-0.5} [1 - (1-\delta) \{1 - \exp(-0.8)\}]$$

[0.5]

The credit spread on a bond is difference between the yield on the bond and yield on similar government bond, which we in this case take to be risk free force of interest of 5%.

C_i is the credit spread on bond for company i and r_i is the continuously compounded yield on the bond.

$$\begin{aligned} C_A &= r_A - 0.05 \\ 2C_B &= r_A - 0.05 \\ r_A &= 2C_B + 0.05 \end{aligned}$$

$$\begin{aligned} C_B &= r_B - 0.05 \\ r_B &= C_B + 0.05 \end{aligned}$$

[1]

we can express the price of zero coupon bond in terms of continuously compounded yield on the bond:

$$\begin{aligned} B_A &= \exp(-10 r_A) = \exp(-10 (2C_B + 0.05)) \\ B_B &= \exp(-10 r_B) = \exp(-10 (C_B + 0.05)) \end{aligned}$$

[1]

This gives simultaneous equations:

$$\begin{aligned} \exp(-10 (2C_B + 0.05)) &= e^{-0.5} [1 - (1-\delta) \{1 - \exp(-2.5)\}] \\ \exp(-10 (C_B + 0.05)) &= e^{-0.5} [1 - (1-\delta) \{1 - \exp(-0.8)\}] \end{aligned}$$

Cancelling $e^{-0.5}$ gives:

$$\begin{aligned} \exp(-20 C_B) &= 1 - (1-\delta) \{1 - \exp(-2.5)\} \\ \exp(-10 C_B) &= 1 - (1-\delta) \{1 - \exp(-0.8)\} \end{aligned}$$

[1]

Squaring equation 1 and substituting it into equation 2:

$$[1 - (1-\delta) \{1 - \exp(-0.8)\}]^2 = 1 - (1-\delta) \{1 - \exp(-2.5)\}$$

$$1 - 2(1-\delta) \{1 - \exp(-0.8)\} + (1-\delta)^2 (1 - e^{-0.8})^2 = 1 - (1-\delta) \{1 - \exp(-2.5)\}$$

$$-2(1-\delta) \{1 - \exp(-0.8)\} + (1-\delta)^2 (1 - e^{-0.8})^2 = - (1-\delta) \{1 - \exp(-2.5)\}$$

Cancelling $(1-\delta)$ on both sides:

$$-2(1 - e^{-0.8}) + (1-\delta)(1 - e^{-0.8})^2 = - (1 - \exp(-2.5))$$

$$(1-\delta) = [2(1 - e^{-0.8}) - (1 - e^{-2.5})] / (1 - e^{-0.8})^2$$

$$1-\delta = 0.6049$$

$$\delta = 0.3951 \text{ (or 39.51\%)}$$

[2]

[7]

[11 Marks]

Solution 7:

- i) Strictly positive means that $A(t) > 0$ for all time t .

[0.5]

The supermartingale property means that, whenever $t < T$:

$$E_P[A(T)/\mathcal{F}_t] \leq A(t)$$

[1]

The presence of the exponential function ensures that this function is strictly positive

[0.5]

Since $A(t)$ is strictly positive, the supermartingale property is equivalent to:

$$E_P[A(T)/F_t] / A(t) \leq 1$$

$$\text{L.H.S} = e^{-.08t - 0.02B(t)} E_P[e^{-.08T + 0.02B(T)} / F_t]$$

$$= e^{-.08(T-t)} E_P[e^{0.02[B(T) - B(t)]} / F_t]$$

[1.5]

Because of the independent property of Brownian motion, we can drop the F_t . We can then use the fact that $B(T) - B(t) \sim N(0, T-t)$ under P to evaluate the expectation.

$$= e^{-.08(T-t)} e^{1/2 * (.02)^2 * (T-t)}$$

$$= e^{-.0798(T-t)}$$

When $t < T$, this is indeed less than 1, so the supermartingale property is satisfied.

[1.5]

[5]

ii) Answer is (b)

[2]

iii) Answer is (a)

[2]

$$A(t) = e^{-.08t + 0.02B(t)}$$

$$A(t) = e^{X(t)}$$

$$dA(t) = e^{X(t)} dX(t) + \frac{1}{2} e^{X(t)} [dX(t)]^2$$

$$dA(t) = A(t) [dX(t) + \frac{1}{2} [dX(t)]^2]$$

$$X(t) = -.08t + 0.02B(t)$$

$$dX(t) = -.08dt + 0.02dB(t)$$

$$[dX(t)]^2 = .02^2 dt$$

$$dA(t) = A(t) [-.08dt + 0.02dB(t) + \frac{1}{2} .02^2 dt]$$

$$dA(t) = A(t) [-.0798dt + 0.02dB(t)]$$

the drift and volatility coefficients are

$$\mu_A(t) = -.0798 \text{ and } \sigma_A(t) = 0.02$$

iv) Answer is (d)

[2]

$$P(t, T) = E_P[A(T)/F_t] / A(t) = e^{-.0798(T-t)}$$

$$P(2, 12) = e^{-.798} = 0.4502$$

v) Answer is (a)

[2]

The price of a 5% annual coupon bond per 100 nominal is:

$$P = 5 [v + v^2 + \dots + v^{10}] + 100v^{10}$$

$$\text{Where } v = e^{-.0798} = 0.9233$$

Evaluating the sum as a geometric progression, we get:

$$P = 5v [(1 - v^{10}) / (1 - v)] + 100v^{10}$$

$$P = 5v [(1-v^{10})/(1-v)] + 100v^{10}$$

$$P = 5 * .9233 * [(1-.450224)/(1-0.9233)] + 100 * 0.450224$$

$$P = 78.11$$

[13 Marks]

Solution 8:

i) C

[2]

Reason: Inflation adjusted chain-ladder method allows inflation explicitly

ii)

Development factors = 1.1327, 1.1315, 1.0304

[1.5]

Cumulative DFs = 1.3207, 1.1660, 1.0304

[1.5]

Ultimate loss ratio = 4,400/4,500 = 0.9777

[0.5]

Estimated ultimate loss for each accident year:

For 2017 :

$$0.9777 * 4,900 = 4,791.1$$

For 2018 :

$$0.9777 * 5,050 = 4,937.8$$

For 2019:

$$0.9777 * 5,200 = 5084.4$$

[1.5]

Revised unexpired losses are:

For 2017 :

$$(1-1/1.0304) * 4,791.1 = 141.555$$

For 2018 :

$$(1-1/1.1660) * 4,937.8 = 702.959$$

For 2019 :

$$(1-1/1.3207) * 5084.4 = 1234.735$$

[1.5]

Revised ultimate losses are:

For 2016 : 4400

For 2017 :

$$141.555 + 4590 = 4732$$

For 2018 :

$$702.959 + 4290 = 4993$$

For 2019 :

$$1234.735 + 4150 = 5385$$

[1.5]

Total = 19509.25

[0.5]

Claims till date = 1.7 crore = 17000 [in thousands]

Reserve required = 2509.25 [in thousands]

[0.5]

[9]

[11 Marks]

Solution 9:

i)

$$R(w) = w * u'' / u'$$

[1]

$$u' = w^{\lambda-1}, u'' = (\lambda-1) * w^{\lambda-2}$$

[1]

$$R(W) = -(\lambda - 1)$$

if $R(W) = 0$, $-(\lambda - 1) = 0 \Rightarrow \lambda = 1$

[1]

[3]

ii)

Let P be the Max Premium that Uday can pay

At $\lambda = 1$, $U = w - 1$

$U(w - P) = E(U(w - X))$, where w = wealth, X = loss amount

Since $U = w - 1$, Utility function is almost equal to wealth.

[1]

Therefore, expected loss of wealth would equal loss of utility

$$P = 3/10 * 30000 = 9000$$

[1]

[2]

iii)

Let Q be the Min Premium that Insurer can offer

At $\lambda = 1$, $U = w - 1$

$U(w - P) = E(U(w - X))$, where w = wealth, X = loss amount

Since $U = w - 1$, Utility function is almost equal to wealth.

[1]

Therefore, expected loss of wealth would equal loss of utility

$$\Rightarrow Q = \text{prob of loss} * \text{claim size} = 3/10 * 30000 = 9000$$

[1]

[2]

iv)

Let P be the Max Premium that uday can pay

At $\lambda = .9$, $U = (w^{.9} - 1)/.9$

[0.5]

$U(w - P) = E(U(w - X))$, where w = wealth, X = loss amount

[0.5]

$$U(w - P) = ((50000 - P)^{.9} - 1)/.9 \quad \text{----- eq 1}$$

[0.5]

$$E(U(w - X)) = ((50000 - 30000)^{.9} - 1)/.9 * .3 + ((50000)^{.9} - 1)/.9 * .7$$

[1]

$$= 15655.60135 \quad \text{----- eq2}$$

[1]

Equating Eq 1 with eq2, gives

$$50000 - P = 40732.1$$

[2]

(candidates need not write every interim step)

$$\Rightarrow P = 9267.904$$

[0.5]

[6]

v)

$U''(w) < 0$ for $\lambda < 1$. Therefore, The Insurer has become risk-averse. The minimum premium would have increased as compared to value obtained in (c)

[1]

[14 Marks]

Solution 10:

i)

C_t : American call

Min value of American call is the same as min value of European call because it is never optimal to exercise an American call before its expiry for a non-dividend-paying stock. So, from put-call parity (or otherwise), the min value is $S - K \cdot \exp(-rt) = 20 - 22 \cdot \exp(-0.12 \cdot 1) = 0.49$

Max value of American call is the value of the stock, i.e. $S = 20$

p_t : European put

Min value of European put from put-call parity (or otherwise) is $K \cdot \exp(-rt) - S = -0.49$. However, since an option value cannot be less negative, the lower bound here is zero.

Max value of European put is the discounted value of the strike price $= K \cdot \exp(-rt) = 19.51$.

P_t : American put

Min value of American put is its intrinsic value $= K - S = 2$.

Max value of American put is the strike price $= K = 22$.

[1 mark for each bound – includes 0.5 marks for explanation, 0.25 for formula, 0.25 for value]
[6]

ii)

From put-call parity $= c_t = p_t + S - K \exp(-rt) = 10.49$

[2]

iii) C

[2]

iv) In order to maintain a constant portfolio delta, the continuous rebalancing of portfolio is required. It is called dynamic delta hedging.

[1]

v) A financial market is said to be complete if any contingent claim (i.e. derivative payoff) can be replicated using the underlying asset and cash.

[1]

It is important for derivatives to have complete market specially over the counter derivatives as the price of derivative can be determined using replicating portfolio of the underlying asset's market value and cash value. If market is not complete, the replicating portfolio is not possible to build.

[1]

[2]

vi)

- Decrease in underlying share price **increases** put option price
- Increase in dividend rate **increases** put option price
- Reduction in time to expiry **decreases** put option price
- Increase in strike price **increases** put option price
- Reduction in risk-free interest rate **increases** put option price
- Increase in volatility of share price **increases** put option price

[0.5 each]

[3]

[16 Marks]
